01 / PROBLEM AND CONTRIBUTION

Speech analysis with FFT & STFT

Digital Signal Processing Lab mini-project | Undergraduate | Two-person team

RESULT AT A GLANCE

Produced waveform and spectrogram views using a 64-sample Hamming window, 32-sample overlap, and 64-point transform, making the analysis configuration explicit.

The engineering question

How does the frequency content of a speech sample change over time?

Situation

A single spectrum cannot show when different frequency components occur in a speech sample. This study compares the audio waveform and FFT view with a time-frequency representation.

Task

Use MATLAB to read the sample, calculate the FFT, and inspect local frequency content using the supplied STFT configuration.

My contribution

Collaborative MATLAB speech-analysis project with B. K. V. Surya.

Tools and concepts

MATLAB / FFT / STFT / Windowing / Signal visualization

02 / DESIGN AND METHOD

From the task to implementation

01 / Implementation step

Read the audio samples and sampling rate with audioread, inspected the waveform, and calculated an FFT magnitude.

02 / Implementation step

Used a 64-sample Hamming window, 32-sample overlap, and a 64-point transform in the spectrogram calculation to inspect local frequency content.

Windowed analysis

The STFT uses a 64-sample Hamming window with 32 samples of overlap and a 64-point transform. It produces a local time-frequency view, whereas the full FFT combines information from the entire supplied signal.

Architecture / workflow

Conceptual summary of the documented workflow; not a simulation result.

01	Audio sample
02	FFT magnitude
03	Windowed STFT
04	Spectrogram

03 / RESULTS AND ENGINEERING PERSPECTIVE

Evidence, interpretation, and next steps

STFT window	64 samples
Overlap	32 samples
FFT length	64 points

Reported result

The report contains waveform and spectrogram output; the ZIP includes the MATLAB script used for the analysis.

Axis calibration matters

The supplied FFT plotting code scales its index using the sampling rate directly. Correct frequency-bin scaling is needed before reading numeric frequencies from that plot. The spectrogram configuration is reported without inferring recognition or noise-reduction performance.

Scope and limitations

The time-frequency view is the strongest demonstration in this study. Before extending the analysis, the saved FFT plot's frequency-axis scaling needs correction so that frequency readings are calibrated.

Possible next steps - proposed, not completed results

- Correct the FFT frequency axis and label waveform time units consistently.
- Compare several window lengths on a known signal to explain time/frequency resolution.

Source material: Ruthvik_Surya_DSP_MiniProject.pdf and DSP/DSP_Miniproject.m.

Prepared from the supplied reports, project files, and verified portfolio copy. Figures retain their original content; explanatory diagrams are labelled.

Project files and documentation: Speech-Signal-Processing-DSP